Research Article

Received: 19 February 2026 Published online in Wiley Online Library:

(wileyonlinelibrary.com) DOI 10.1002/jsfa.70681

# Black flesh disorder in mango: a chilling injury symptom increased by early harvest and low storage temperature and reduced by 1-methylcyclopropene

**Bruna Parente de Carvalho Pires,[a] João Claudio Vilvert,[a] Nilo Ricardo Corrêa de Mello Júnior,[b] Sandy Raiele Sena Monteiro,[c] Mikaele de Souza Santos,[c] Ester Silva Regis,[c] Jackson Teixeira Lobo,[d] Jeffrey K. Brecht[e] and Sergio Tonetto de Freitas[a*]**

## Abstract

**BACKGROUND: Black flesh (BF) is an internal disorder in mango, characterized by the development of dark brown to black pigmentation in the inner mesocarp tissue during storage or transport. This study investigated the effects of harvest maturity, low storage or low transport temperatures, and 1-methylcyclopropene (1-MCP) on mango fruit susceptibility to BF. In the first experiment, 'Tommy Atkins', 'Kent', and 'Palmer' mangoes were harvested at two maturity stages and BF incidence was assessed after 30 days' storage at 8, 10, or 12.5 °C plus 7 days of shelf life at 20 °C. In the second experiment, 'Tommy Atkins' and 'Keitt' mangoes were also treated at harvest with the ethylene action inhibitor 1-MCP (0–800 nL $L^{-1}$) and stored at 9 °C for 45 days, followed by 7 days of shelf life at 20 °C. The most effective 1-MCP treatment concentration (200 nL $L^{-1}$) was then validated on mangoes for reduction of BF incidence and severity.**

**RESULTS: The incidence of BF increased with early harvest and lower storage temperatures (≤10 °C), but it was not observed at 12.5 °C. 1-Methylcyclopropene at 200 nL $L^{-1}$ markedly reduced the incidence and severity of BF, while suppressing respiration and ethylene production. This concentration also maintained flesh firmness and antioxidant capacity, performing in a similar way to higher concentrations.**

**CONCLUSION: Early harvest and low storage temperature increased the incidence of BF whereas 1-MCP treatment reduced it.**
**© 2026 The Author(s). *Journal of the Science of Food and Agriculture* published by John Wiley & Sons Ltd on behalf of Society of Chemical Industry.**

**Keywords:** *Mangifera indica*; 1-MCP; chilling injury; internal disorder; antioxidant activity; phenolic compounds

## INTRODUCTION

Mango (*Mangifera indica* L.), which is cultivated in tropical and subtropical regions, is commercially important worldwide due to its pleasant flavor and aroma, exotic appearance, and high nutritional value.[1,2] Brazil ranks as the sixth largest producer of mangoes worldwide, with an estimated production of 1.76 million t in 2023.[3] The expansion of mango cultivation in the country has been driven by its high commercial value, especially in the international market, with strong demand for cultivars such as 'Tommy Atkins', 'Palmer', 'Keitt', and 'Kent' in Europe, Asia, and North America.[2,4,5,6]

Mango is currently Brazil's primary fruit export, with the irrigated semi-arid São Francisco Valley accounting for over 90% of national exports.[7] Shipping mangoes to distant markets requires low temperatures to manage their rapid ripening.[8] However,

* Correspondence to: ST Freitas, Brazilian Agricultural Research Corporation, Tropical Semi-Arid Embrapa, 56302-970, Petrolina, PE, Brazil. E-mail: sergio.freitas@embrapa.br

a Brazilian Agricultural Research Corporation, Tropical Semi-Arid Embrapa, Petrolina, Brazil

b Faculty of Agricultural and Veterinary Sciences, São Paulo State University, Jaboticabal, Brazil

c Graduate Program in Agronomy, Federal University of Vale do São Francisco, Petrolina, Brazil

d AgroFresh Brasil Ltda, São Paulo, Brazil

e Horticultural Sciences Department, Institute of Food and Agricultural Sciences, University of Florida, Gainesville, FL, USA

© 2026 The Author(s). *Journal of the Science of Food and Agriculture* published by John Wiley & Sons Ltd on behalf of Society of Chemical Industry. This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

mangoes are highly sensitive to cold, which can lead to chilling injury (CI).[9] Commonly reported CI symptoms include aroma loss and uneven ripening, lenticel discoloration, and peel browning.[10,11] Recent findings suggest that low temperatures may trigger internal flesh darkening,[12] but no further studies have tested the effects of cold temperatures on the incidence and severity of this condition, referred to as 'black flesh' (BF).

Black flesh, which is also known as cutting black, black spot, flesh browning, *corte negro*, or *pulpa negra*, is an internal disorder of mango fruit characterized by a diffuse flesh discoloration around the seed, ranging from brown to black, possibly resulting from cell wall and membrane degradation.[12-14] It has been recently reported to be one of the most important mango internal disorders, which can affect up to 100% of the fruit during storage and/or shipping under commercial conditions, representing a major concern for the mango industry.[12,15]

Given reports linking BF incidence to low storage temperatures,[12] it is important to understand the mechanisms regulating CI in order to identify factors contributing to BF. A primary CI mechanism is the reduction of the unsaturated-to-saturated fatty acid ratio in cellular membranes. This increases membrane rigidity and susceptibility to lipid peroxidation, triggered by reactive oxygen species (ROS) under cold stress. This process is exacerbated by early harvest and the exposure of immature fruit to low temperatures.[16]

1-Methylcyclopropene (1-MCP) – a gaseous cyclopropene derivative that irreversibly binds to cellular ethylene receptors[1]1 – has emerged as a potential technology for mitigating CI. The effects of 1-MCP have been extensively investigated for extending shelf life and delaying senescence.[17–19] Recent studies have shown that 1-MCP has not only inhibited ethylene-induced ripening processes, but has also improved membrane stability by scavenging ROS.[11,20–22]

Reduced CI symptoms following 1-MCP application have been documented in avocado,[23] nectarine,[24] and apple.[25] In mangoes,[26] the mitigation of external CI symptoms by 1-MCP has been demonstrated in two cultivars. Studies have reported that 1-MCP-treated mangoes exhibited lower malondialdehyde (MDA) content – a byproduct of lipid peroxidation – suggesting enhanced membrane integrity via reduced oxidative stress.[27] 1-Methylcyclopropene therefore increased fruit resistance to chilling stress, potentially playing a role in preventing the internal browning associated with BF. However, no studies have specifically evaluated whether BF represents a form of CI or tested the effectiveness of 1-MCP in mitigating the disorder. Based on this evidence, it was hypothesized that BF incidence could be increased in mango fruit using factors that increase fruit susceptibility to CI, such as early harvest and low storage temperature, and reduced with treatments that reduce susceptibility, such as 1-MCP treatment.

The aims of this study were therefore to: (1) investigate the relationship between maturity stage, storage temperature, and BF incidence in mangoes; (2) determine the optimal 1-MCP concentration for alleviating BF; and (3) validate the effectiveness of 1-MCP in controlling BF while evaluating its influence on physiological attributes, quality parameters, and antioxidant metabolism. The storage durations employed (28–45 days) are consistent with commercial mango international transport, storage, and distribution times.

## MATERIALS AND METHODS

### Plant material and experimental design

The experiments were conducted with mangoes harvested from commercial orchards in the São Francisco Valley, Petrolina, PE, Brazil. The orchards were selected based on their history of high BF incidence. The region has a semi-arid (BSh) climate, according to the Köppen classification.[28] After harvest, fruit were transported to the Postharvest Physiology and Technology Laboratory at Embrapa, Petrolina, PE, Brazil, for treatment application and subsequent assessment. Fruit were selected for uniform size and absence of defects, visible injuries, and diseases.

The study comprised three experimental stages, conducted over 3 years, as summarized in Fig. 1. In the first stage, the incidence and severity of BF were evaluated in mangoes harvested at two maturity stages (early and tree ripe) and stored at three temperatures, to determine the effect of harvest maturity and storage temperature on BF incidence and severity, a possible CI disorder. In the second stage, different postharvest 1-MCP (AgroFresh Brasil Ltda, São Paulo, Brazil) doses were tested

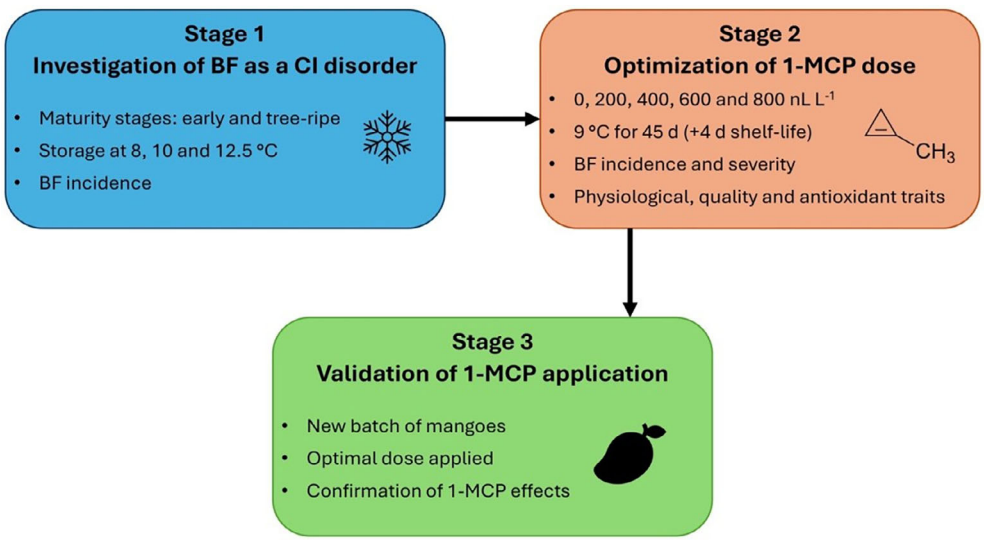


**Figure 1.** Overview of the experimental stages of the study.



 © 2026 The Author(s). 
*Journal of the Science of Food and Agriculture* published by John Wiley & Sons Ltd on behalf of Society of Chemical Industry.

10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

to alleviate BF in early maturity stage mangoes and assess their effects on fruit quality and antioxidant metabolism, with the goal of identifying an optimal dose. In the third stage, the optimal 1-MCP dose was applied to a new batch of early maturity stage mangoes as a validation step. The three stages are described in detail below.

### Experiment 1: effect of maturity stage and storage temperature on BF incidence

A total of 600 'Tommy Atkins', 600 'Kent', and 600 'Palmer' mangoes were harvested at two maturity stages, corresponding to stage 1 (early maturity; i.e. physiologically mature but prior to onset of ripening) and Stage 3 (advanced, 'tree-ripe' maturity; approximately one-half ripe based on internal color development) on a standard commercial mango 1–5 maturity/ripeness scale.[29] At the early maturity stage, 'Tommy Atkins' mangoes showed flesh firmness of 84.2 N, soluble solid content (SSC) of 10.7%, and titratable acidity (TA) of 1.03%. At this stage, 'Palmer' mangoes had a firmness of 115.9 N, SSC of 10.3%, and TA of 1.0%, whereas 'Kent' mangoes showed flesh firmness of 89.4 N, SSC of 9.6%, and TA of 1.1%. At the advanced maturity stage, 'Tommy Atkins' mangoes showed flesh firmness of 47.0 N, SSC of 12.6%, and TA of 0.67%, whereas 'Palmer' mangoes had a firmness of 100.8 N, SSC of 12.3%, and TA of 0.69%, and 'Kent' mangoes showed flesh firmness of 56.9 N, SSC of 12.3%, and TA of 0.58%.

Fruit from each cultivar and maturity stage were stored at 8 °C, 10 °C, or 12.5 °C with relative humidity of 90% to 95% for 30 days. These temperatures have been reported to result in high, medium, and low mango susceptibility to external CI symptoms during storage.[14,30,31] A previous report also suggested that storage temperatures above 10 °C may reduce BF incidence in mangoes.[12] Each treatment (cultivar × maturity stage × temperature) consisted of 100 fruit, divided into four replications of 25 fruit each. After cold storage, fruit were kept at 20 °C for 7 days to simulate shelf life conditions, at which point BF incidence was assessed. Incidence was calculated as the percentage of fruit with visual BF symptoms in each sample.

### Experiment 2: effect of 1-MCP application on BF and fruit quality

A total of 540 'Tommy Atkins' and 540 'Keitt' mangoes were harvested at earlier maturity stages (1,2).[29] A subsample of 20 fruit per cultivar was analyzed at harvest to determine initial physicochemical quality. The remaining fruit were randomly assigned to five treatment groups, with four replicates of 104 fruit each.

The treatments were 0, 200, 400, 600, and 800 nL $L^{-1}$ 1-methylcyclopropene (1-MCP) (AgroFresh Brasil Ltda, São Paulo, Brazil) (SmartFresh). Fruit were placed in sealed 120 L containers, and the appropriate amounts of 1-MCP were dissolved in 25 mL of water and added to the containers for 24 h gas exposure at 9 °C. Following treatment, fruit were stored at 9 °C with relative humidity of 90% to 95% for 45 days. After this period, half of the fruit were analyzed immediately and the other half were held at 20 °C for 7 days to simulate shelf life. After both periods, fruit were evaluated for BF incidence and severity, physiological traits (respiration, ethylene production, and flesh firmness), physicochemical traits (peel and flesh color, soluble solids, and titratable acidity), and antioxidant metabolism (phenolic compounds and antioxidant activity by 2,2′-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) (Sigma-Aldrich Brasil Ltda., São Paulo, Brazil) (ABTS) and 2,2-diphenyl-1-picrylhydrazyl (Sigma-Aldrich Brasil Ltda., São Paulo, Brazil) (DPPH) assays), as described below.

#### *Black flesh incidence and severity*

Fruit were sliced longitudinally on both sides, parallel to and near the flat side of the stone, to expose the flesh to evaluate BF incidence and severity. Incidence (%) was calculated as described above. Severity (%) was assessed visually only in BF-affected fruit by estimating the percentage of flesh area exhibiting symptoms. These values were then averaged across all symptomatic fruit within each sample.

#### *Physiological parameters*

Respiration rate and ethylene production were measured using a portable gas analyzer (model F-950; Felix Instruments, WA, USA) in a static system with a 1 h chamber sealing period. Flesh firmness was determined as the bioyield point with a texture analyzer (model TA.XTplus; Stable Micro Systems, Godalming, UK) fitted with a 6 mm cylindrical probe used to penetrate the fullest area of the cheek after removal of a small patch of peel. Firmness results were expressed in N as the force required to penetrate 10 mm into the flesh.

#### *Physicochemical quality parameters*

Peel and flesh color were measured with a digital colorimeter (model CR-400; Minolta, Tokyo, Japan) using the CIE L*a*b* color space and expressed as hue angle (0/360° = red, 90° = yellow, 180° = green, and 270° = blue). The peel ground color was measured while avoiding areas with red blush. Flesh color was measured on the inner side of each fruit half. Measurements were obtained by slicing the fruit longitudinally, parallel and close to the flat surface of the stone. The SSC was measured in juice extracted from the flesh using a handheld digital refractometer (model PAL-1; Atago, Tokyo, Japan) and expressed as a percentage. The TA was determined from 1.0 g of expressed juice diluted in 50 mL of distilled water, which was titrated with 0.1mol $L^{-1}$ NaOH to pH 8.1, using an automatic titrator (model Titrino Plus 848; Metrohm, São Paulo, Brazil), and expressed as percentage citric acid equivalents.

#### *Total phenolic content and antioxidant activity*

Total phenolic content (TPC) was determined using the Folin–Ciocâlteu method, based on the reducing capacity of phenolic compounds. For extract preparation, 8.0 g of flesh was extracted with 20 mL of 50% (Sigma-Aldrich Brasil Ltda., São Paulo, Brazil) methanol (v/v) for 1 h, centrifuged at 10 000 × *g* for 15 min at 4 °C, and filtered. The residue was then extracted with 20 mL of 70% acetone (v/v) for 1 h under the same conditions. Both supernatants were combined in a 50 mL volumetric flask and adjusted with distilled water. The extract was used for TPC and antioxidant assays, as described below.

Aliquots of 0.5 mL of extract were diluted with distilled water to 1.0 mL, then mixed with 1.0 mL (Sigma-Aldrich Brasil Ltda., São Paulo, Brazil) Folin–Ciocâlteu reagent, 2.0 mL of 20% (Sigma-Aldrich Brasil Ltda., São Paulo, Brazil) sodium carbonate, and 2.0 mL distilled water. After 30 min, absorbance was measured at 700 nm. Results were expressed as mg 100 $g^{-1}$ GAE fresh weight.[32]

Antioxidant activity (AOX) was assessed using the ABTS and DPPH radical scavenging assays.[33,34] In the ABTS assay, the $ABTS^{\bullet+}$ radical cation was generated by mixing equal volumes of 7 mM ABTS and 2.45 mM (Sigma-Aldrich Brasil Ltda., São Paulo, Brazil)


© 2026 The Author(s).


10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

potassium persulfate, kept in dark conditions for 16 h at 20 °C. Before use, the solution was diluted with ethanol to an absorbance of 0.700 at 754 nm. For the assay, 30 μL of extract was added to 3.0 mL of ABTS solution. Absorbance was read at 0 and 6 min at 754 nm. Results were expressed as mM (Sigma-Aldrich Brasil Ltda., São Paulo, Brazil) Trolox 100 $g^{-1}$.

In the DPPH assay, a 100 μM DPPH stock solution was prepared in methanol and 0.1 mL of extract was added to 2.9 mL of 1.0 mM DPPH solution. After 30 min in dark conditions at 20 °C, absorbance was read at 517 nm. Results were expressed as mM Trolox 100 $g^{-1}$.

### Experiment 3: validation of the optimal 1-MCP dose

After evaluating and selecting the most efficient 1-MCP dose (200 nL $L^{-1}$) to control BF incidence and severity, stage 1–2 'Tommy Atkins' and 'Keitt' mangoes were harvested again to validate the results. In the validation stage, only control (0 nL $L^{-1}$) and 1-MCP treatment (200 nL $L^{-1}$) were repeated as in Experiment 2 (see above) for each cultivar. Each treatment consisted of four replications containing 12 fruit per replication. The fruit were stored and evaluated after 45 days of storage and 7 days of shelf life, as described above.

### Statistical analysis

Experiments 1 and 2 followed a completely randomized design (CRD) with four replications per treatment. A one-way analysis of variance (ANOVA) was carried out in five steps: (1) detection of extreme outliers in the dataset; (2) assessment of normality assumption through model residuals using Q–Q plots and the Shapiro–Wilk test; (3) evaluation of homogeneity of variances by residuals versus fitted plots and Levene's test; (4) application of the F-test to assess treatment effects; and (5) pairwise mean comparisons among treatments using the least significant difference (LSD) test ($P < 0.05$). In Experiment 3, treatment comparisons (1-MCP at 0 × 200 nL $L^{-1}$) were performed using non-parametric bootstrap resampling. Bootstrap analysis was carried out with 5000 iterations, resampling with replacement from the original dataset. The mean difference between the control and treatment was calculated for each iteration. The distribution of bootstrap estimates was then used to obtain percentile-based 95% confidence intervals for the mean difference.

## RESULTS

### Experiment 1: BF incidence in response to early harvest and storage temperature

For the three mango cultivars evaluated in Experiment 1, BF incidence was influenced by fruit maturity at harvest and by storage temperature ($P < 0.05$) (Fig. 2). For 'Tommy Atkins' mangoes, fruit harvested at the early maturity stage exhibited a markedly higher BF incidence, particularly at lower storage temperatures, reaching 70% at 8 °C and 67% at 10 °C (Fig. 2). Conversely, no symptoms were observed at 12.5 °C, regardless of maturity stage (Fig. 2). Fruit harvested at the more advanced maturity showed a substantial reduction in BF incidence at the same temperatures, with values decreasing to 22% at 8 °C and 11% at 10 °C (Fig. 2).

For 'Kent' mangoes, BF symptoms were also associated strongly with low storage temperatures and harvest maturity (Fig. 2). Early harvested fruit exhibited high incidences of BF, reaching 80% at 8 °C and 78% at 10 °C, whereas no symptoms were detected at 12.5 °C (Fig. 2). Fruit harvested at the more advanced maturity showed markedly lower BF incidence, which remained at 33% at both 8 °C and 10 °C, and no symptoms were observed at 12.5 °C (Fig. 2).

For 'Palmer' mangoes, early harvested fruit stored at 8 °C showed the highest BF incidence (89%), which decreased to 33% at 10 °C (Fig. 2). Although late-harvested 'Palmer' fruit still exhibited symptoms at 8 °C and 10 °C (22% and 32%, respectively), their incidence remained lower than in early harvested fruit stored at 8 °C (Fig. 2). For both maturity stages, no symptoms were detected at 12.5 °C (Fig. 2).

### Experiment 2: BF incidence and severity in response to 1-MCP treatment

*Incidence and severity of BF*

Treatment with 1-MCP significantly ($P < 0.05$) reduced the BF incidence and severity in both 'Tommy Atkins' and 'Keitt' mangoes stored at 9 °C for 45 days (Figs 3 and 4). At the end of cold storage, untreated 'Tommy Atkins' fruit exhibited 100% BF incidence, whereas the application of 1-MCP (200–800 nL $L^{-1}$) decreased the disorder to below 60% ($P < 0.05$). After 7 days of shelf life at 20 °C, untreated 'Tommy Atkins' fruit also showed higher BF incidence than fruit treated with 200 nL $L^{-1}$ and 600 nL $L^{-1}$ 1-MCP-treated fruit (Fig. 3). After 45 days of storage, untreated 'Tommy Atkins' fruit had the highest BF severity (51%), compared with fruit treated with 1-MCP at 200, 600, and 800 nL $L^{-1}$ (which displayed BF severity of 17% to 26%). After 7 days of shelf life, untreated 'Tommy Atkins' fruit maintained higher BF severity (53%) than 1-MCP-treated fruit at 600 nL $L^{-1}$ (26%) (Fig. 3).

For 'Keitt' mangoes, the BF incidence reached 100% in untreated fruit after 45 days of cold storage, whereas 1-MCP treatment, regardless of the concentration, decreased the disorder to below 60%. The application of 1-MCP at 600 nL $L^{-1}$ promoted the lowest BF incidence (27%) after cold storage. After shelf life, BF incidence was reduced from 100% in untreated fruit to below 65% in all treated fruit (Fig. 4). The severity of BF in 'Keitt' mangoes was also reduced by 1-MCP treatment ($P < 0.05$), both after cold storage and after 7 days of shelf life. Untreated 'Keitt' mangoes exhibited high severity levels (75% after cold storage and 60% after shelf life), whereas 1-MCP-treated fruit showed a drastic reduction, with severity dropping to below 20% after cold storage and from 18% to 20% after shelf life. After storage and shelf life, no significant differences in severity were observed among the 1-MCP doses tested ($P > 0.05$) (Fig. 4).

*Physiological parameters*

Ethylene production was reduced markedly by 1-MCP application in 'Tommy Atkins' and 'Keitt' mangoes, after 45 days of storage and after 7 days of shelf life (Fig. 5(A),(B)). After storage, 'Tommy Atkins' untreated fruit produced ethylene at a rate of 2.18 nL $kg^{-1}$ $h^{-1}$, whereas fruit treated with 1-MCP at 400 nL $L^{-1}$ (0.10 nL $kg^{-1}$ $h^{-1}$) or 600 nL $L^{-1}$ (0.38 nL $kg^{-1}$ $h^{-1}$) exhibited lower ethylene production (Fig. 5(A)). For 'Keitt', untreated fruit produced ethylene at 11.58 nL $kg^{-1}$ $h^{-1}$, whereas 1-MCP-treated fruit ethylene production ranged from 0.00 to 0.35 nL $kg^{-1}$ $h^{-1}$, with no significant differences between 1-MCP doses (Fig. 5(B)).

After 7 days of shelf life, ethylene production decreased from 9.52 nL $kg^{-1}$ $h^{-1}$ for 'Tommy Atkins' untreated fruit to 1.05–1.80 nL $kg^{-1}$ $h^{-1}$ in 1-MCP-treated fruit (Fig. 5(A)). A similar pattern was observed for 'Keitt' mangoes, with untreated fruit producing 19.13 nL $kg^{-1}$ $h^{-1}$ of ethylene compared with 0.13–2.18 nL $kg^{-1}$ $h^{-1}$ for 1-MCP-treated fruit (Fig. 5(B)).

The respiration rate of both cultivars was affected significantly by 1-MCP application. Untreated 'Tommy Atkins' mangoes had


© 2026 The Author(s).


10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

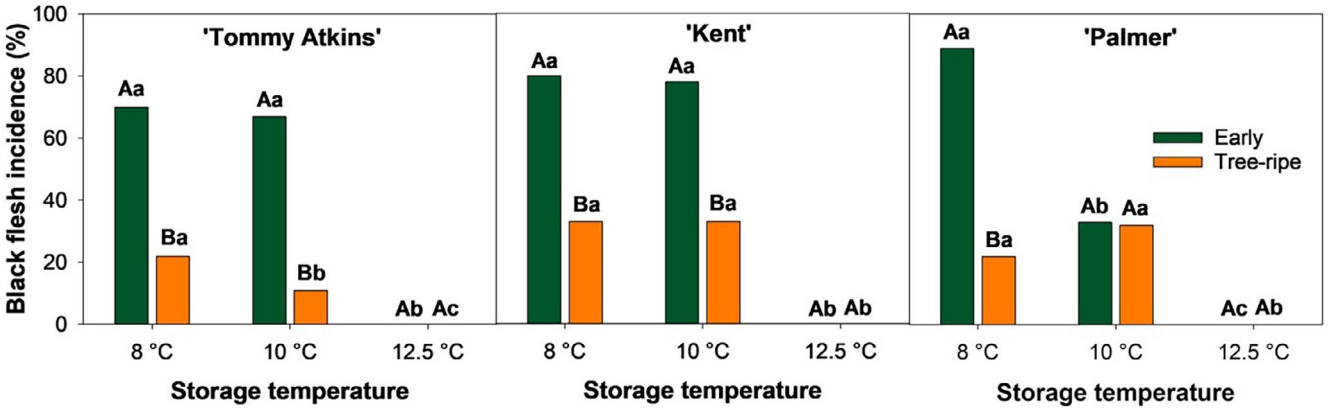


**Figure 2.** Incidence of black flesh (BF) in 'Tommy Atkins', 'Kent', and 'Palmer' mangoes in response to different maturity stage and storage temperature. The BF incidence was determined after 30 days of cold storage plus 7 days at 20 °C. Within each storage temperature, means followed by the same uppercase letter – and within each maturity stage by the same lowercase letter – are not significantly different according to the least significant difference (LSD) test ($P < 0.05$).

the highest respiration rate after cold storage (22.94 mL $CO_2$ $kg^{-1}$ $h^{-1}$), and 1-MCP administered at 200, 400, and 600 nL $L^{-1}$ reduced respiration to rates ranging from 16.17 to 17.45 mL $CO_2$ $kg^{-1}$ $h^{-1}$. After 7 days of shelf life, respiration was higher in untreated fruit (51.32 mL $CO_2$ $kg^{-1}$ $h^{-1}$) than in 1-MCP-treated fruit (43.50–49.90 mL $CO_2$ $kg^{-1}$ $h^{-1}$) (Fig. 5(C)).

A similar pattern was observed with 'Keitt' mangoes. After cold storage, untreated fruit had a respiration rate of 23.07 mL $CO_2$ $kg^{-1}$ $h^{-1}$, whereas 1-MCP-treated fruit maintained lower rates (12.30–17.67 mL $CO_2$ $kg^{-1}$ $h^{-1}$). After shelf life, respiration reached 46.64 mL $CO_2$ $kg^{-1}$ $h^{-1}$ in untreated fruit, whereas 1-MCP application reduced respiration to 30.68–35.09 mL $CO_2$ $kg^{-1}$ $h^{-1}$ (Fig. 5(D)).

The flesh firmness of both cultivars was significantly better maintained by 1-MCP application. For 'Tommy Atkins' mangoes, after cold storage, untreated fruit exhibited the lowest firmness (12.09 N), whereas 1-MCP-treated fruit retained significantly higher values, of 48.31–55.32 N, with no significant differences among doses. After 7 days of shelf life, firmness of untreated fruit decreased to 5.43 N, whereas 1-MCP-treated fruit maintained higher flesh firmness values of 7.78–9.18 N (Fig. 5(E)).

A similar response was observed for 'Keitt' mangoes. After cold storage, untreated fruit showed firmness of 30.27 N, while 1-MCP-treated fruit retained flesh firmness values of 43.12–47.26 N. After shelf life, untreated fruit softened drastically to 10.48 N, whereas 1-MCP-treated fruit maintained significantly higher firmness of 23.65–27.88 N (Fig. 5(F)).

*Color and physicochemical attributes*

The peel color, expressed as hue angle, was not significantly affected by 1-MCP treatments in either 'Tommy Atkins' or 'Keitt' mangoes after cold storage. For 'Tommy Atkins', peel hue angle was 82.2–87.5° (Fig. 6(A)), whereas for 'Keitt' it was 107.3–110.3° (Fig. 6(B)). After shelf life, 'Tommy Atkins' peel hue values were not affected by 1-MCP, at 74.0–78.3° (Fig. 6(A)), whereas for 'Keitt', untreated fruit had a lower hue (95.2°), compared with 1-MCP-treated fruit (105.5–107.4°) (Fig. 6(B)).

Flesh color was affected significantly by 1-MCP application for both 'Tommy Atkins' and 'Keitt' mangoes. After cold storage, untreated fruit exhibited lower hue angle values than 1-MCP-treated fruit, indicating more advanced ripeness in both untreated cultivars. In contrast, 1-MCP-treated fruit maintained greater hue angles, with values of 77.4–78.4° for 'Tommy Atkins' (Fig. 6(C)) and 79.6–80.3° for 'Keitt' mangoes (Fig. 6(D)), regardless of dose. After 7 days of shelf life, flesh color differences between untreated and 1-MCP-treated fruit became more pronounced. For 'Tommy Atkins', hue angle decreased to 72.9° in untreated fruit, and in 1-MCP-treated fruit it was 75.8–76.4° (Fig. 6(C)). A similar pattern was observed for 'Keitt', for which untreated fruit reached lower hue values (67.0°) compared with 1-MCP-treated fruit (79.4–80.4°) (Fig. 6(D)).

The SSC was not consistently affected by 1-MCP application in either cultivar. For 'Tommy Atkins', SSC values remained stable across all treatments, ranging from 11.5% to 12.9%, both after cold storage and after 7 days of shelf life (Fig. 6(E)). For 'Keitt', a slight but significant reduction in SSC was observed after cold storage in fruit treated with the intermediate 1-MCP dose (400 nL $L^{-1}$), but this effect was not maintained after shelf life, when all treatments showed similar values (13.0% to 14.3%) (Fig. 6(F)).

The application of 1-MCP had no effect on TA in 'Tommy Atkins' mangoes, regardless of whether the fruit were assessed after cold storage or after 7 days of shelf life. After cold storage, TA values ranged from 0.68 to 0.94%, and after shelf life, TA values were reduced to 0.39% and 0.63%, with no differences among treatments (Fig. 6(G)).

In contrast, significant 1-MCP effects were observed on the TA of 'Keitt' mangoes. After cold storage, TA ranged from 0.76% in untreated fruit to 0.95% or 0.96% in fruit treated with 400 or 600 nL $L^{-1}$ 1-MCP, respectively, with intermediate values in the other treatments. After shelf life, the differences became more pronounced, with untreated fruit showing the highest acidity (0.52%), while 1-MCP-treated fruit had lower values, ranging from 0.33% to 0.40% (Fig. 6(H)).

*Total phenolic content and AOX*

The TPC was influenced significantly by 1-MCP application for both 'Tommy Atkins' and 'Keitt' mangoes. At harvest, TPC averaged 46.9 mg GAE 100 $g^{-1}$ for 'Tommy Atkins'. After 45 days of cold storage, untreated fruit exhibited the lowest TPC content (8.6 mg GAE 100 $g^{-1}$), whereas 1-MCP-treated fruit maintained markedly higher levels, especially with the 600 nL $L^{-1}$ dose (78.6 mg GAE 100 $g^{-1}$). These differences were consistent after the 7 days of shelf life, when untreated fruit showed a very low


© 2026 The Author(s).
Journal of the Science of Food and Agriculture published by John Wiley & Sons Ltd on behalf of Society of Chemical Industry.

10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

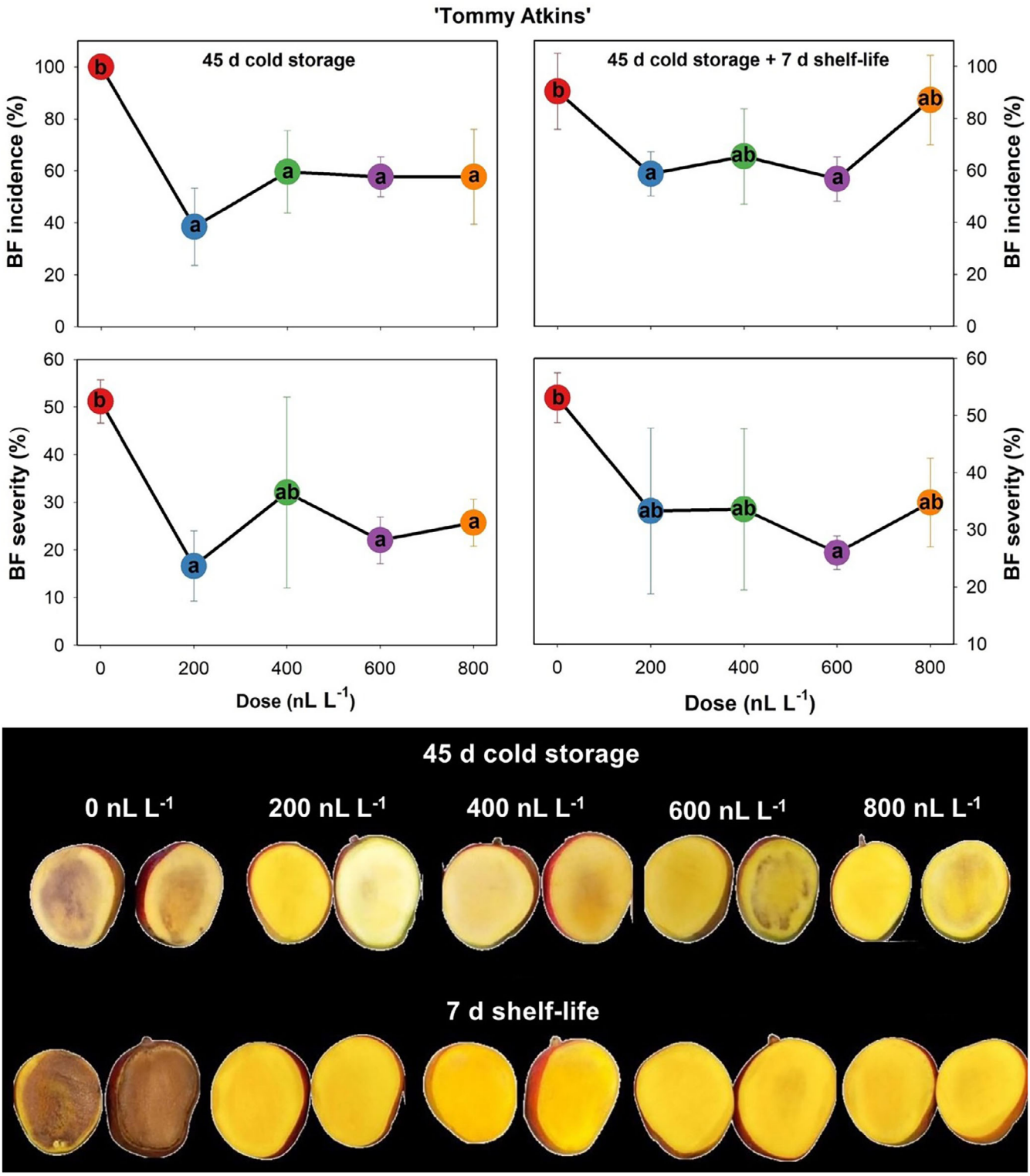


**Figure 3.** Incidence and severity of black flesh (BF) disorder in 'Tommy Atkins' mangoes treated with different 1-methylcyclopropene (1-MCP) concentrations and stored at 9 °C for 45 days followed by 7 days of shelf life at 20 °C. Data are presented as means ± standard deviations (SDs). Different letters indicate significant differences among 1-MCP doses according to the least significant difference (LSD) test ($P < 0.05$).

TPC content (8.6 mg GAE 100 $g^{-1}$), compared with 1-MCP treatments. Among 1-MCP doses, 200 nL $L^{-1}$ (69.2 mg GAE 100 $g^{-1}$) and 400 nL $L^{-1}$ (73.7 mg GAE 100 $g^{-1}$) retained higher TPC values (Fig. 7(A)).

For 'Keitt', the average TPC content at harvest was 38.6 mg GAE 100 $g^{-1}$. After 45 days of cold storage, untreated fruit showed the lowest values (9.4 mg GAE 100 $g^{-1}$), whereas 1-MCP-treated fruit maintained higher TPC content, ranging from 27.4 to 32.2 mg



 © 2026 The Author(s). 
*Journal of the Science of Food and Agriculture* published by John Wiley & Sons Ltd on behalf of Society of Chemical Industry.

10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

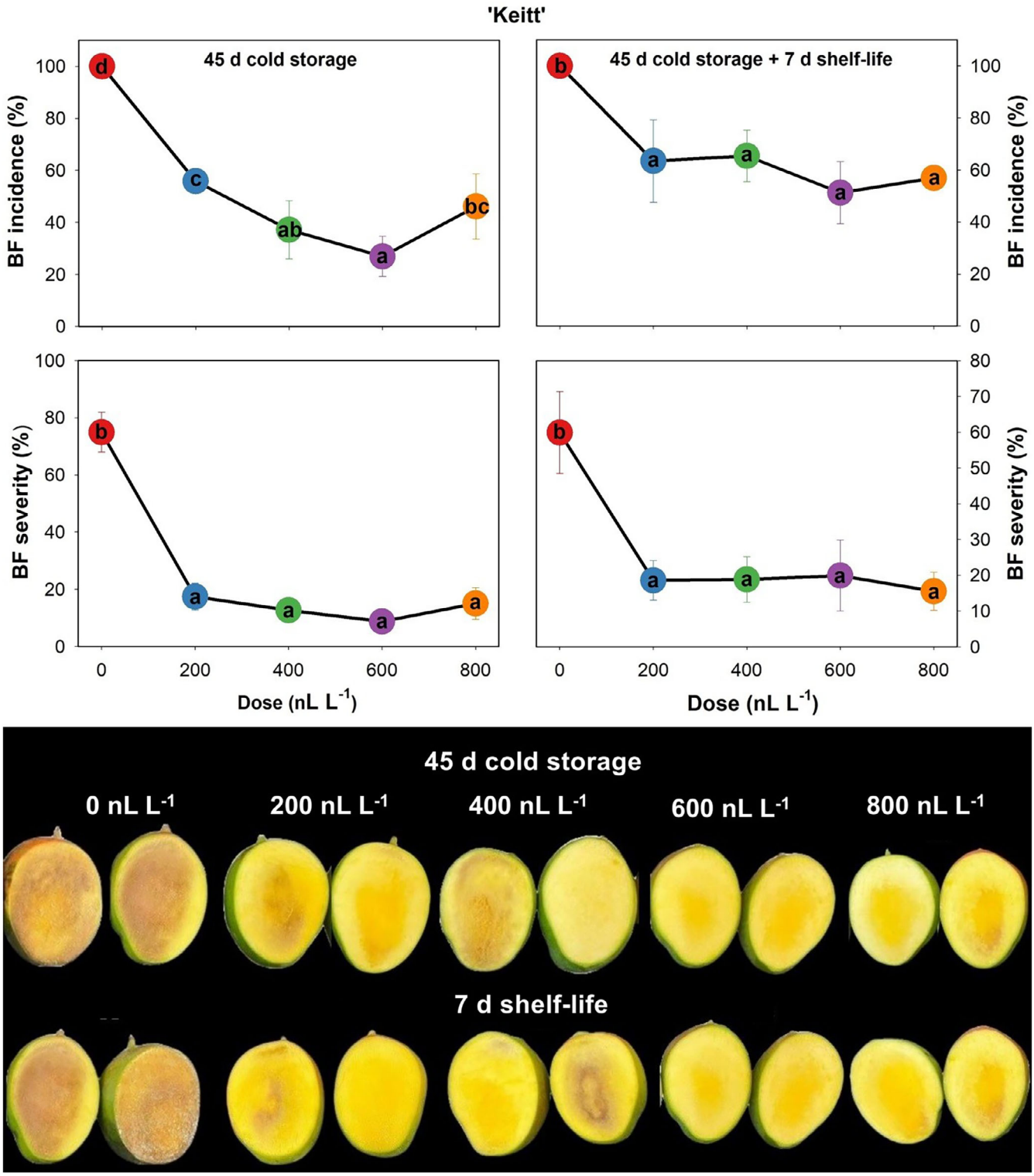


**Figure 4.** Incidence and severity of black flesh (BF) disorder in 'Keitt' mangoes treated with different 1-methylcyclopropene (1-MCP) concentrations and stored at 9 °C for 45 days followed by 7 days of shelf life at 20 °C. Data are presented as means ± standard deviations (SDs). Different letters indicate significant differences among 1-MCP doses according to the least significant difference (LSD) test ($P < 0.05$).

GAE 100 $g^{-1}$. The highest TPC levels were observed with 1-MCP doses of 800 and 200 nL $L^{-1}$ (32.2 and 31.7 mg GAE 100 $g^{-1}$, respectively). After 7 days of shelf life, untreated fruit presented lower TPC content (9.4 mg GAE 100 $g^{-1}$), whereas all 1-MCP treatments preserved significantly higher TPC content (23.8–29.6 mg GAE 100 $g^{-1}$), without significant differences among the 1-MCP doses ($P > 0.05$) (Fig. 7(B)).

The low TPC content in untreated fruit was reflected in lower AOX values measured by both the ABTS and DPPH assays ($P < 0.05$). For the ABTS method and 'Tommy Atkins', AOX was



10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

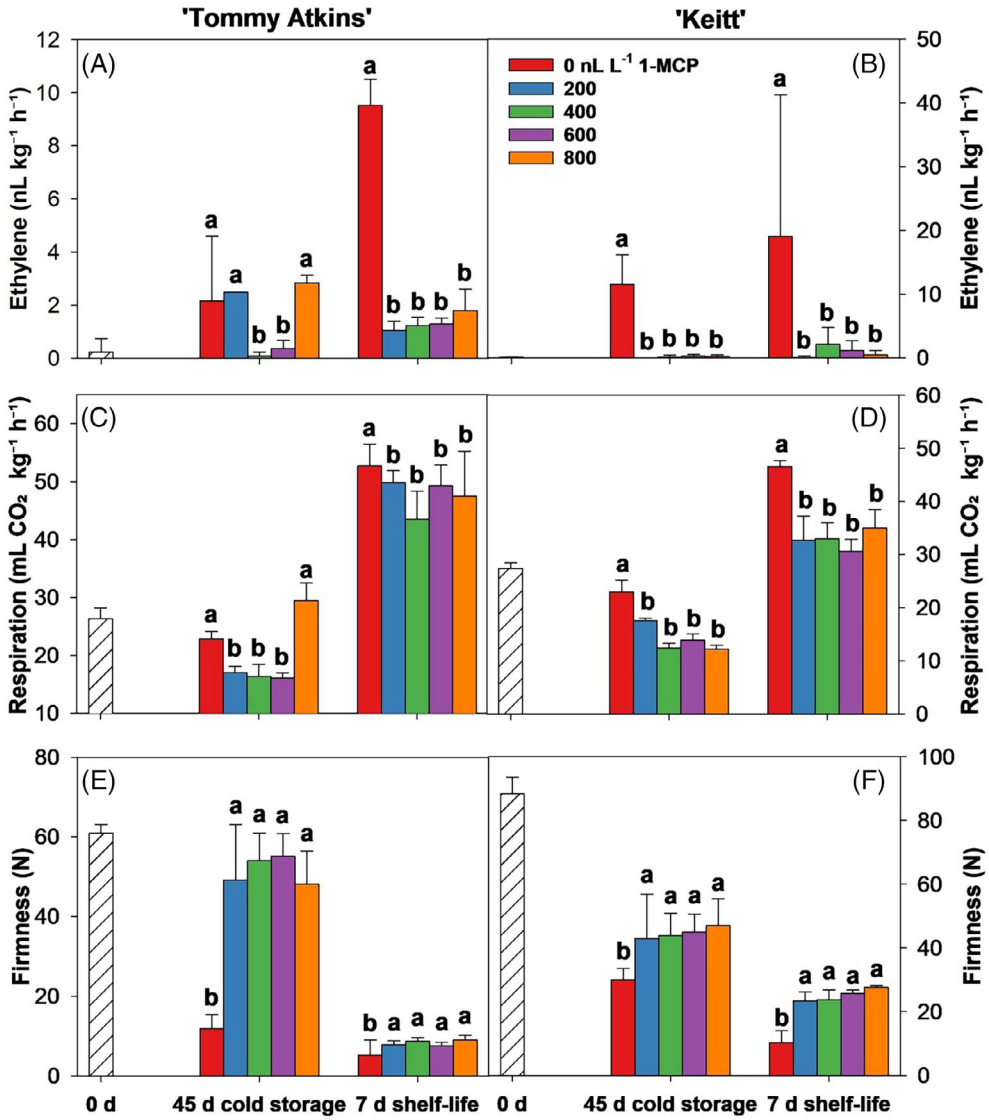


**Figure 5.** Effects of 1-methylcyclopropene (1-MCP) on ethylene production in 'Tommy Atkins' (A) and 'Keitt' (B), respiration rate in 'Tommy Atkins' (C) and 'Keitt' (D), and flesh firmness in 'Tommy Atkins' (E) and 'Keitt' (F) mangoes stored at 9 °C for 45 days followed by 7 days of shelf life at 20 °C. Data are presented as means ± standard deviations (SDs). Different letters indicate significant differences among 1-MCP doses according to the least significant difference (LSD) test ($P < 0.05$).

9.21 mM Trolox 100 $g^{-1}$ at harvest. After cold storage, the AOX in untreated fruit was reduced to 2.70 mM Trolox 100 $g^{-1}$, and 1-MCP-treated fruit retained higher AOX, especially at 600 nL $L^{-1}$ (10.03 mM Trolox 100 $g^{-1}$). After shelf life, untreated fruit had lower AOX (2.50 mM Trolox 100 $g^{-1}$) than 1-MCP-treated fruit, especially at 600 nL $L^{-1}$ (6.23 mM Trolox 100 $g^{-1}$) (Fig. 7(C)).

A similar pattern was observed with 'Keitt' mango. The AOX was 8.14 mM Trolox 100 $g^{-1}$ at harvest, decreasing to 1.43 mM Trolox 100 $g^{-1}$ in untreated fruit after cold storage, whereas the 1-MCP-treated fruit retained a higher AOX (3.66–5.24 mM Trolox 100 $g^{-1}$). After shelf life, untreated fruit had very low AOX (1.36 mM Trolox 100 $g^{-1}$), differing significantly from all 1-MCP treatments (2.56–3.24 mg TE 100 $g^{-1}$), except the 600 nL $L^{-1}$ dose (Fig. 7(D)).

For the DPPH method, 'Tommy Atkins' mangoes showed an AOX of 9.75 mg TE 100 $g^{-1}$ at harvest. After 45 days of cold storage, the highest AOX levels were found in 1-MCP-treated fruit (7.85–8.99 mg TE 100 $g^{-1}$), with no differences among doses, whereas untreated fruit showed a significantly lower value (4.68 mg TE 100 $g^{-1}$). After shelf life, 1-MCP at 400 nL $L^{-1}$ resulted in the highest AOX values (8.64 mg TE 100 $g^{-1}$), but did not differ from 600 nL $L^{-1}$, and untreated fruit showed the lowest AOX (4.31 mg TE 100 $g^{-1}$) (Fig. 7(E)).

For 'Keitt', a similar trend was observed with an AOX of 9.10 mg TE 100 $g^{-1}$ at harvest. After 45 days of cold storage, 1-MCP-treated fruit exhibited the highest AOX values (7.17–8.52 mg TE 100 $g^{-1}$), and untreated fruit had the lowest values (3.67 mg TE 100 $g^{-1}$). After shelf life, 1-MCP-treated fruit had the highest AOX (5.92–7.35 mg TE 100 $g^{-1}$), with no differences among doses, whereas untreated fruit had the lowest AOX (3.67 mg TE 100 $g^{-1}$) (Fig. 7(F)).



 © 2026 The Author(s). 
*Journal of the Science of Food and Agriculture* published by John Wiley & Sons Ltd on behalf of Society of Chemical Industry.

10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

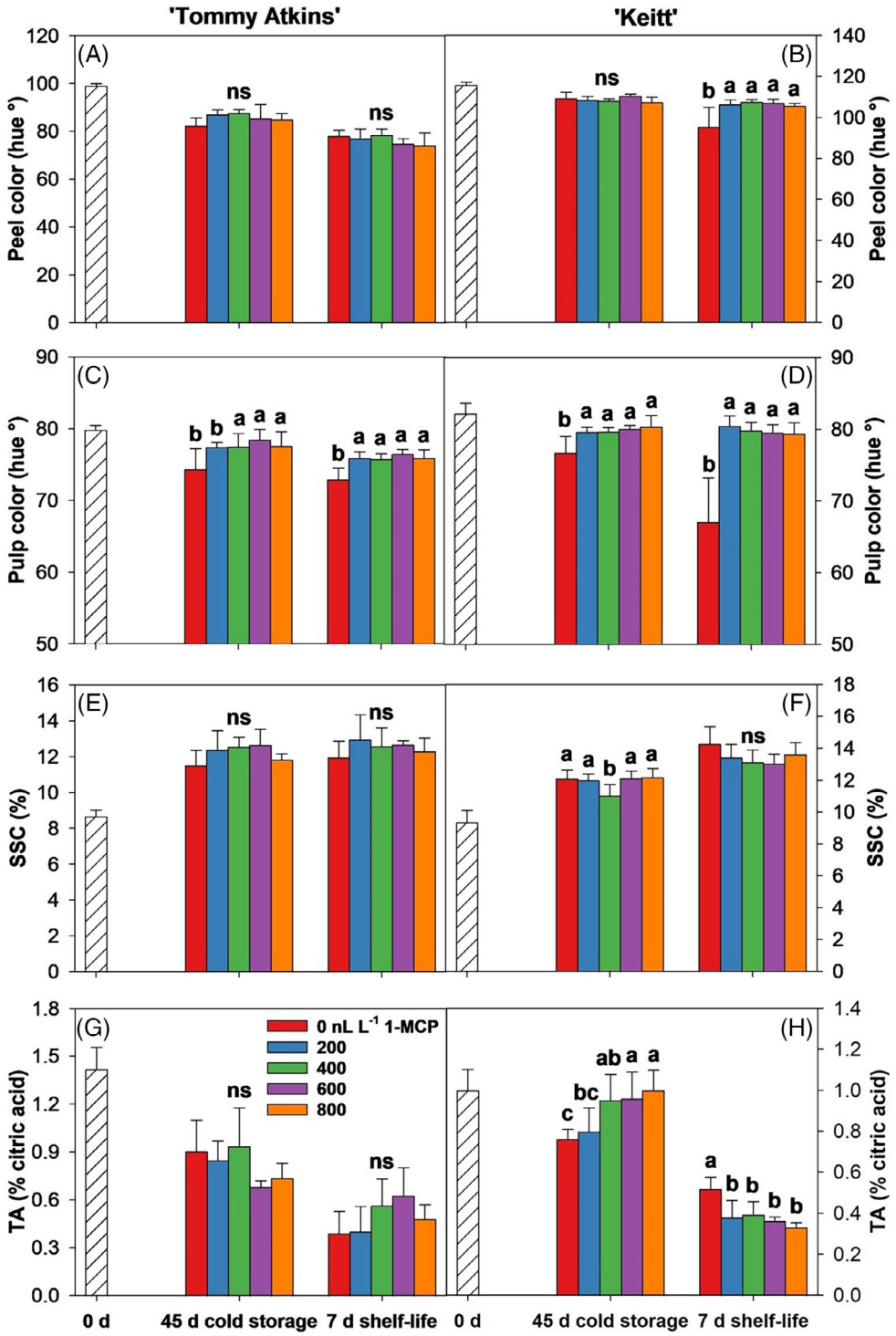


**Figure 6.** Effects of 1-methylcyclopropene (1-MCP) on peel color in 'Tommy Atkins' (A) and 'Keitt' (B), flesh (pulp) color in 'Tommy Atkins' (C) and 'Keitt' (D), soluble solids content (SSC) in 'Tommy Atkins' (E) and 'Keitt' (F), and titratable acidity (TA) in 'Tommy Atkins' (G) and 'Keitt' (H) mangoes stored at 9 °C for 45 days followed by 7 days of shelf life at 20 °C. Data are presented as means ± standard deviations (SDs). Different letters indicate significant differences among 1-MCP doses according to the least significant difference (LSD) test ($P < 0.05$).

### Experiment 3: validation of 1-MCP for alleviating BF and preserving mango quality

According to the study described above, the 1-MCP concentration of 200 nL $L^{-1}$ was the lowest effective dose to inhibit BF incidence and reduce its severity, showing similar BF inhibition with higher doses. The 1-MCP dose of 200 nL $L^{-1}$ was therefore chosen to be validated in a new batch of 'Tommy Atkins' and 'Keitt' mangoes produced in the São Francisco Valley, Petrolina, PE, Brazil the following year.

For 'Tommy Atkins' mangoes stored at 9 °C for 45 days, 1-MCP treatment reduced BF incidence from 45% to 20%, while severity decreased from 8.4% to 6.7% (Table 1). Following 7 days of shelf


© 2026 The Author(s).


10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

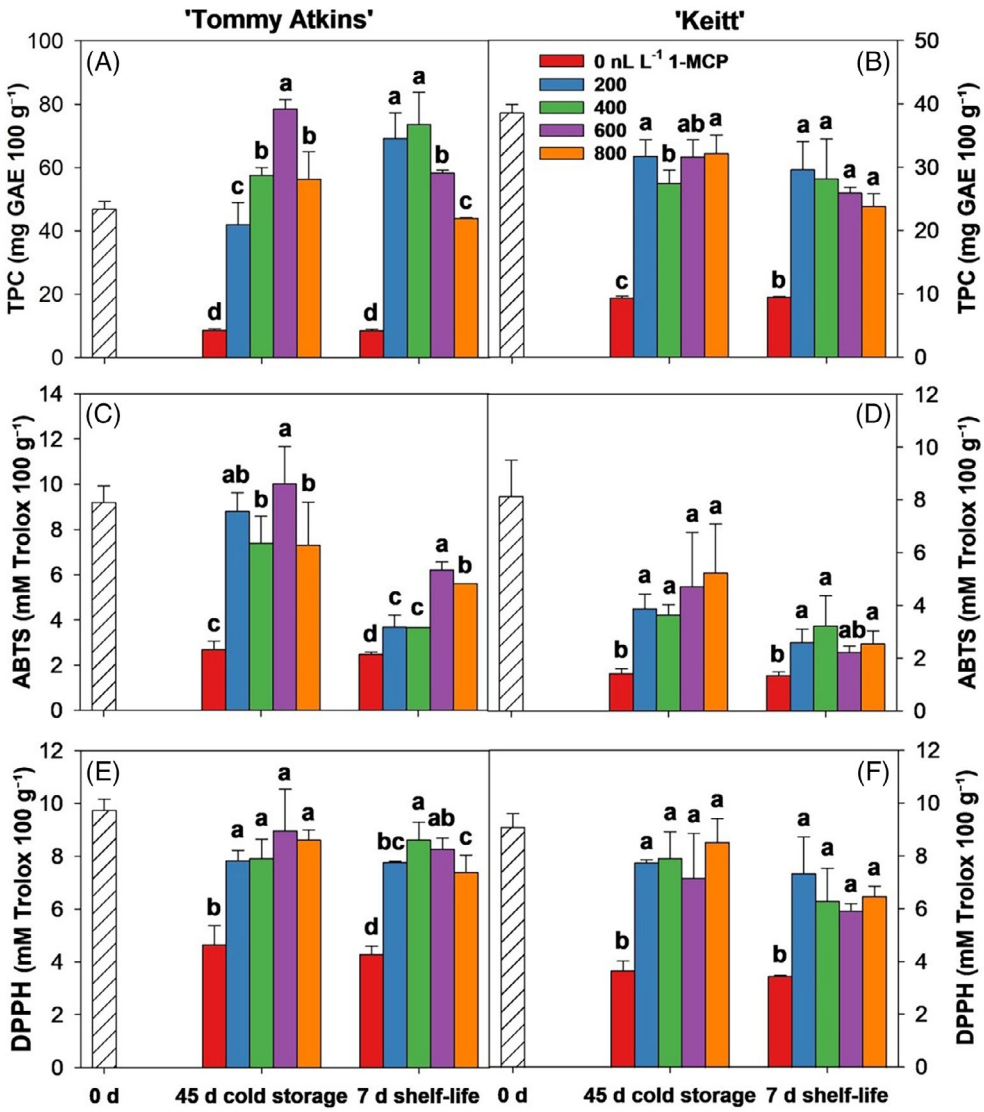


**Figure 7.** Effects of 1-MCP on total phenolic content (TPC) in 'Tommy Atkins' (A) and 'Keitt' (B), and antioxidant activity determined by the ABTS method in 'Tommy Atkins' (C) and 'Keitt' (D) and the DPPH method in 'Tommy Atkins' (E) and 'Keitt' (F) mangoes stored at 9 °C for 45 days followed by 7 days of shelf life at 20 °C. Data are presented as means ± standard deviations (SDs). Different letters indicate significant differences among 1-MCP doses according to the least significant difference (LSD) test ($P < 0.05$).

life, incidence was 26.6% in treated fruit compared to 66.6% in the control; similarly, severity was reduced to 7.0% from 36.2% (Table 1).

For 'Keitt' mangoes, the impact was more pronounced. After 45 days at 9 °C, 1-MCP reduced BF incidence from 100% to 2.5% and severity from 72.5% to 3.7% (Table 1). After shelf life, treated fruit maintained a low incidence (6.2%) and severity (1.8%), whereas untreated fruit remained at 100% incidence and 83.6% severity (Table 1).

The application of 1-MCP markedly suppressed ethylene production in both cultivars (Table 1). For 'Tommy Atkins', ethylene production by untreated fruit was 0.22 nL $kg^{-1}$ $h^{-1}$ after 45 days of cold storage, and treated fruit ethylene production was undetectable. After shelf life, ethylene production by untreated fruit increased to 6.10 nL $kg^{-1}$ $h^{-1}$, whereas treated fruit exhibited low ethylene production (0.50 nL $kg^{-1}$ $h^{-1}$). A similar response was observed with 'Keitt' mangoes, for which 1-MCP reduced ethylene production from 15.50 to 0.10 nL $kg^{-1}$ $h^{-1}$ after storage, and from 18.40 to 0.70 nL $kg^{-1}$ $h^{-1}$ after shelf life (Table 1).

The respiration rate was also significantly reduced by 1-MCP. For 'Tommy Atkins', respiration was decreased by 1-MCP from 21.5 to 9.1 mL $CO_2$ $kg^{-1}$ $h^{-1}$ after cold storage and from 54.5 to 45.9 mL $CO_2$ $kg^{-1}$ $h^{-1}$ after shelf life. For 'Keitt', untreated fruit exhibited high respiration rates of 36.2 and 62.5 mL $CO_2$ $kg^{-1}$ $h^{-1}$, whereas 1-MCP-treated fruit showed significantly lower rates of 23.4 and 46.8 mL $CO_2$ $kg^{-1}$ $h^{-1}$ after storage and shelf life, respectively (Table 1).

Fruit firmness was significantly preserved by 1-MCP application ($P < 0.05$). 'Tommy Atkins' mangoes had an initial firmness of 55.4 N at harvest. After cold storage, 1-MCP-treated fruit remained firmer (30.5 N) than untreated fruit (25.5 N), and this difference persisted after shelf life with values of 14.2 N for treated fruit and 7.7 N for untreated fruit. The effect was even more pronounced in 'Keitt', for which untreated fruit softened from



 © 2026 The Author(s). 
*Journal of the Science of Food and Agriculture* published by John Wiley & Sons Ltd on behalf of Society of Chemical Industry.

10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**Table 1.** Black flesh (BF) incidence and severity, and physiological attributes of untreated (0 nL $L^{-1}$) and 1-methylcyclopropene (1-MCP)-treated (200 nL $L^{-1}$) 'Tommy Atkins' and 'Keitt' mangoes after 45 days of storage at 9 °C and 7 days of shelf life at 20 °C

| | | 45 days of cold storage | | 45 days plus 7 days shelf life | |
|---|---|---|---|---|---|
| Variable | At harvest | Control | 1-MCP | Control | 1-MCP |
| Tommy Atkins | | | | | |
| BF incidence (% of fruit) | 0 ± 0 | 45.0 ± 5.17 a* | 20.0 ± 7.18 b | 66.6 ± 18.86 a | 26.6 ± 9.43 b |
| BF severity (% of flesh) | 0 ± 0 | 8.4 ± 5.22 a | 6.7 ± 2.22 b | 36.2 ± 15.44 a | 7.0 ± 6.00 b |
| Ethylene (nL $kg^{-1}$ $h^{-1}$) | 0.02 ± 0.05 | 0.22 ± 0.12 a | 0.00 ± 0.00 b | 6.10 ± 4.70 a | 0.50 ± 0.78 b |
| Respiration (mL $CO_2$ $kg^{-1}$ $h^{-1}$) | 14.3 ± 1.70 | 21.5 ± 1.10 a | 9.1 ± 0.60 b | 54.5 ± 3.0 a | 45.9 ± 0.9 b |
| Firmness (*N*) | 55.4 ± 4.30 | 25.5 ± 2.50 b | 30.5 ± 0.50 a | 7.7 ± 3.7 b | 14.2 ± 2.5 a |
| Peel color (°hue) | 98.1 ± 3.30 | 83.5 ± 0.60 a | 83.1 ± 2.30 a | 82.6 ± 3.4 a | 82.6 ± 3.5 a |
| Flesh color (°hue) | 76.5 ± 1.80 | 77.3 ± 0.82 a | 74.7 ± 0.60 b | 75.3 ± 1.2 a | 75.5 ± 0.7 a |
| SSC (%) | 8.5 ± 0.40 | 10.3 ± 0.90 b | 11.7 ± 0.40 a | 10.0 ± 2.0 a | 11.4 ± 0.6 a |
| TA (% citric acid) | 1.80 ± 0.08 | 1.15 ± 0.03 b | 1.23 ± 0.07 a | 0.57 ± 0.19 a | 0.65 ± 0.09 a |
| TPC (mg GAE 100 $g^{-1}$ FW) | 48.23 ± 0.66 | 22.76 ± 0.92 b | 39.99 ± 1.19 a | 7.97 ± 0.48 b | 20.78 ± 0.72 a |
| ABTS (mg TE 100 $g^{-1}$ FW) | 10.18 ± 0.31 | 4.64 ± 0.73 b | 8.23 ± 0.29 a | 3.47 ± 0.32 b | 6.48 ± 0.11 a |
| DPPH (mg TE 100 $g^{-1}$ FW) | 11.45 ± 0.40 | 6.23 ± 0.38 b | 9.35 ± 0.43 a | 4.87 ± 0.35 b | 7.90 ± 0.16 a |
| Keitt | | | | | |
| BF incidence (% of fruit) | 0 ± 0 | 100.0 ± 0.00 a | 2.5 ± 2.00 b | 100.0 ± 0.00 a | 6.2 ± 5.22 b |
| BF severity (% of flesh) | 0 ± 0 | 72.5 ± 7.72 a | 3.7 ± 3.50 b | 83.6 ± 13.14 a | 1.8 ± 1.20 b |
| Ethylene (nL $kg^{-1}$ $h^{-1}$) | 0.15 ± 0.06 | 15.50 ± 2.29 a | 0.10 ± 0.08 b | 18.40 ± 7.62 a | 0.70 ± 0.36 b |
| Respiration (mL $CO_2$ $kg^{-1}$ $h^{-1}$) | 21.3 ± 1.85 | 36.2 ± 2.80 a | 23.4 ± 5.50 b | 62.5 ± 8.9 a | 46.8 ± 5.0 b |
| Firmness (N) | 98.2 ± 8.40 | 2.2 ± 0.10 b | 42.3 ± 16.70 a | 2.3 ± 0.3 b | 11.7 ± 4.5 a |
| Peel color (°hue) | 103.4 ± 3.90 | 104.6 ± 5.70 b | 111.5 ± 2.80 a | 96.7 ± 5.1 b | 108.2 ± 2.0 a |
| Flesh color (°hue) | 81.3 ± 2.80 | 72.1 ± 5.00 b | 78.0 ± 1.10 a | 63.3 ± 5.2 b | 77.1 ± 1.1 a |
| SSC (%) | 8.2 ± 1.70 | 13.7 ± 0.90 a | 13.5 ± 0.60 a | 12.2 ± 0.5 b | 14.0 ± 0.6 a |
| TA (% citric acid) | 1.36 ± 0.11 | 0.96 ± 0.03 b | 1.09 ± 0.04 a | 0.88 ± 0.13 a | 0.78 ± 0.13 a |
| TPC (mg GAE 100 $g^{-1}$ FW) | 46.91 ± 1.16 | 16.95 ± 0.36 b | 33.38 ± 0.80 a | 10.53 ± 0.17 b | 16.58 ± 0.17 a |
| ABTS (mg TE 100 $g^{-1}$ FW) | 9.82 ± 0.25 | 2.86 ± 0.52 b | 7.09 ± 0.55 a | 2.24 ± 0.32 b | 5.40 ± 0.22 a |
| DPPH (mg TE 100 $g^{-1}$ FW) | 10.19 ± 0.18 | 6.51 ± 0.40 b | 10.93 ± 0.83 a | 5.04 ± 0.08 b | 7.91 ± 0.18 a |

*Mean values followed by the same letter for each variable are statistically equal according to the least significant difference (LSD) test ($P < 0.05$). 1-MCP = methylcyclopropene, BF = black flesh.

98.2 N at harvest to 2.2 N after cold storage and 2.3 N after shelf life, whereas 1-MCP-treated fruit maintained considerably higher flesh firmness of 42.3 N after cold storage and 11.7 N after shelf life (Table 1).

Peel color was not significantly affected by 1-MCP for 'Tommy Atkins' at both evaluation periods. Conversely, for 'Keitt', 1-MCP-treated fruit maintained higher peel hue values (less yellow or orange), compared with untreated fruit, both after cold storage and shelf life. For flesh color, 1-MCP preserved significantly higher hue values in 'Keitt', particularly after shelf life, compared with untreated fruit. In 'Tommy Atkins', differences in flesh color were less evident, with only a slight reduction observed after cold storage (74.7° in 1-MCP treated fruit vs 77.3° in control fruit) (Table 1).

The SSC increased slightly in 1-MCP-treated 'Tommy Atkins' fruit, compared with untreated fruit after cold storage but no significant differences were observed after shelf life. For 'Keitt', SSC was unaffected by 1-MCP after cold storage, but it was significantly higher in treated fruit after shelf life. The TA showed a modestly higher value in 1-MCP-treated 'Tommy Atkins' fruit after cold storage, compared with untreated fruit (1.23 vs 1.15%), whereas no differences were observed between treated and untreated fruit after shelf life. For 'Keitt', TA was slightly higher in 1-MCP-treated fruit after cold storage, compared with untreated fruit, but declined to similar levels to the control after shelf life (Table 1).

The preservation of bioactive compounds was strongly improved by 1-MCP. At harvest, TPC values were 48.23 mg GAE 100 $g^{-1}$ in 'Tommy Atkins' and 46.91 mg GAE 100 $g^{-1}$ in 'Keitt' mangoes. The TPC values in 'Tommy Atkins' fruit were significantly higher with 1-MCP, compared with control fruit, both after cold storage (39.99 vs 22.76 mg GAE 100 $g^{-1}$) and after shelf life (20.78 vs 7.97 mg GAE 100 $g^{-1}$). Similarly, for 'Keitt', TPC was maintained higher in 1-MCP-treated fruit compared with untreated fruit, both after cold storage and after shelf life (Table 1).

The AOX, measured by both ABTS and DPPH assays, was consistently higher in 1-MCP-treated fruit of both cultivars. For 'Tommy Atkins', AOX determined by the ABTS method was 10.18 mg TE 100 $g^{-1}$ at harvest and was retained at 8.23 mg TE 100 $g^{-1}$ after cold storage and 6.48 mg TE 100 $g^{-1}$ after shelf life in 1-MCP-treated fruit. In contrast, untreated fruit showed a marked reduction in AOX, with values decreasing to 4.64 mg TE 100 $g^{-1}$ after storage and 3.47 mg TE 100 $g^{-1}$ after shelf life. A similar trend was observed for antioxidant activity (AOX) measured by DPPH. From an initial harvest value of 11.45 mg TE 100 $g^{-1}$, AOX decreased after cold storage to 9.35 mg TE 100 $g^{-1}$ in treated fruit compared with 6.23 mg TE 100 $g^{-1}$ in the control. Following shelf life, these values further declined to 7.90 and 4.87 mg TE 100 $g^{-1}$, respectively (Table 1).



10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

For 'Keitt' mangoes, AOX measured by the ABTS method was 9.82 mg TE 100 $g^{-1}$ at harvest. Following cold storage, this value decreased slightly in 1-MCP-treated fruit (to 7.09 mg TE 100 $g^{-1}$) but fell sharply in untreated fruit (to 2.86 mg TE 100 $g^{-1}$). After shelf life, 1-MCP-treated fruit maintained higher AOX (5.40 mg TE 100 $g^{-1}$) than untreated fruit (2.24 mg TE 100 $g^{-1}$). A similar pattern occurred with the DPPH method; from a harvest value of 10.19 mg TE 100 $g^{-1}$, AOX after cold storage was 10.9 mg TE 100 $g^{-1}$ in 1-MCP-treated fruit compared with 6.5 mg TE 100 $g^{-1}$ in the control. After shelf life, these values were 7.91 and 5.04 mg TE 100 $g^{-1}$, respectively (Table 1).

## DISCUSSION

Black flesh (BF) is an internal physiological disorder in mango that remains poorly understood, despite its substantial impact on fruit quality and marketability. It is characterized by diffuse darkening of the fruit mesocarp, ranging from brown to black.[12,13] It is often confused with other internal disorders collectively classified as internal breakdown.[14]

Understanding the factors that contribute to BF incidence is crucial in order to develop effective mitigation strategies. Its subtle nature, with minimal external symptoms, hampers detection during commercial handling and underscores the need for deeper investigation into the mechanisms underlying its incidence and severity.[15] The first reports of BF were documented in Mexico for 'Haden' mangoes cold stored for more than 20 days,[35] and for 'Haden' fruit cold stored for more than 22 days.[36] Recently, it has been suggested that BF could be a symptom of CI. This was based on observing it only after the fruit was stored at a temperature of 10 °C but not at 24 °C.[12]

To date, it has been unclear whether BF should be classified as a CI symptom.[12,13] The results of this study, however, indicate clearly that BF represents a symptom of CI strongly influenced by both storage temperature and harvest maturity. Evaluating three cultivars, 'Tommy Atkins', 'Kent', and 'Palmer', it was possible to determine that early harvested fruit consistently exhibited higher BF incidence than later harvested, half-ripe fruit. This observation is consistent with much research showing that sensitivity to CI decreases as mango fruit mature and ripen.[37–39] This study also showed that BF only occurred when fruit were stored at the lower temperatures of 8 and 10 °C, but no symptoms were observed at 12.5 °C in any cultivar or maturity stage, which is also consistent with many previous reports on effects of storage temperature on mango CI.[12,14] It appears that 12.5 °C may represent a safe threshold for preventing BF under the tested conditions. These findings agree with other studies that have reported BF symptoms in 'Zill' mangoes from China after storage for at least 20 days at 5 °C or 28 days at 10 °C, but not at 14 °C.[30,31]

Recently, visible-near infrared (visible–NIR) spectroscopy and machine learning-based modeling have been proposed as alternatives for predicting the incidence of BF.[40] However, effective strategies to prevent or reduce the BF disorder in mango remain limited. This study proposed, for the first time, the application of 1-MCP as a potential postharvest intervention to reduce both BF incidence and severity in mango. The results demonstrated that 1-MCP application was an effective strategy to reduce both the incidence and severity of BF in cold-stored mangoes, as it has also been reported to inhibit CI symptoms in other species such as avocado,[23] nectarine,[24] and apple.[25] In both 'Tommy Atkins' and 'Keitt', BF incidence reached 100% in untreated fruit after 45 days at 9 °C, while even the lowest 1-MCP dose (200 nL $L^{-1}$) reduced the disorder to below 60%. These findings are consistent with the previously known effects of 1-MCP in alleviating CI-related physiological disorders in mango.[11]

The beneficial effects of 1-MCP are closely linked to its ability to suppress ethylene responses, such as ethylene production and respiration, which are key factors associated with CI susceptibility in banana and peach.[19,41] A meta-analysis of 292 studies on climacteric fruits demonstrated that 1-MCP treatments typically reduce internal ethylene concentration by 89%, ethylene production by 65%, and respiration rate by 25%,[42] reflecting its role as an ethylene receptor antagonist and inhibitor of ethylene-mediated signaling pathways.[43,44] In this study, 1-MCP-treated mangoes exhibited substantially lower ethylene production and respiration rates compared with untreated fruit, regardless of the dose. These findings align with reports on other tropical fruits, where 1-MCP suppressed ripening-related physiological parameters.[45,46] As 1-MCP attenuated both ripening and BF development, the results of the experiments in this study suggest that BF may be associated with the ripening of mangoes exposed to chilling conditions.

The results also showed that 1-MCP effectively delayed the softening of 'Tommy Atkins' and 'Keitt' mangoes, mainly during cold storage. Treated fruit maintained greater firmness in comparison with untreated fruit during storage. This result agrees with other studies using 'Kensington Pride' and 'Tainong' mangoes, both with 1000 nL $L^{-1}$ of 1-MCP.[27,47] The preservation of fruit firmness is likely related to the inhibition of ethylene signal transduction by 1-MCP, which leads to cell wall breakdown.[19] Ethylene stimulates the activity of cell wall-degrading enzymes, including polygalacturonase (PG), pectin methylesterase (PME), and cellulases. In mango, 1-MCP counteracts ethylene-induced softening by modulating the expression of genes such as ethylene response factor (*MiERF*) and polygalacturonase (*MiPG*).[48] By irreversibly binding to ethylene receptors, 1-MCP blocks ethylene perception and signal transduction, thereby reducing the expression of genes involved in cell-wall breakdown.[49] However, both 1-MCP treated and untreated fruit showed softening after transferring to shelf-life conditions, reaching the ready-to-eat developmental stage, which is highly desirable to ensure fruit commercialization and consumption. These results suggest that the effect of 1-MCP in maintaining fruit firmness decreased during shelf life. This is possibly related to higher ethylene receptor turnover and synthesis during ripening at the higher shelf-life temperature, which resulted in greater fruit sensitivity to ethylene, as has also been reported in other studies.[50–53]

As BF is an internal disorder, symptoms are typically restricted to the mesocarp and may not be externally visible.[14] Consequently, changes in peel color are not reliable indicators of the presence of physiological disorders. In the flesh, however, fruit subjected to 1-MCP application exhibited a higher hue angle, compared to untreated fruit, which agrees with the findings of previous studies with 'Tommy Atkins' mango.[54]

In this study, an absence of consistent 1-MCP effects was found for SSC and TA. One of the key effects of physiological disorders on sugar metabolism is altered balance between starch and soluble sugars. Starch accumulates in healthy mangoes during fruit development and is subsequently converted into soluble sugars during ripening. However, physiological disorders can cause excessive premature degradation of starch into sugars, leading to an imbalance that might compromise the fruit taste and texture.[55]

Beyond its effects on the ripening physiology of 'Tommy Atkins' and 'Keitt' mangoes, 1-MCP consistently maintained higher AOX


© 2026 The Author(s).


10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

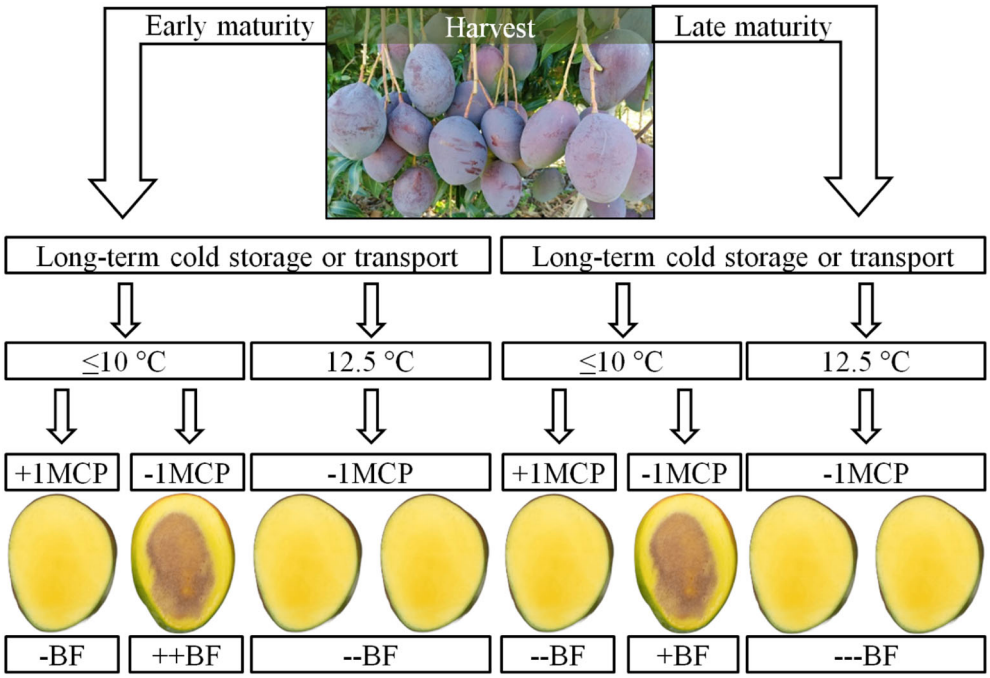


**Figure 8.** Schematic representation illustrating the probability of black flesh (BF) incidence in mangoes during long-term storage or transportation as influenced by harvest maturity, storage temperature, and 1-methylcyclopropene (1-MCP) treatment. Probability of BF incidence was represented as very high (++BF) to very low (−BF).

and TPC in the fruit, compared with untreated fruit. It is known that CI in tropical fruits is closely linked to oxidative stress and the regulation of antioxidant metabolism.[56] Chilling injury-related disorders compromise fruit quality by inducing an imbalance between ROS accumulation and the fruit's antioxidant capacity. Elevated ROS levels, including MDA and hydrogen peroxide ($H_2O_2$), are recognized contributors to CI, as they can impair cell membranes and disrupt metabolic activities that are essential for fruit viability and quality.[49,57] As oxidative stress is a central component of CI, the maintenance of antioxidant metabolism by 1-MCP may play a crucial role in inducing chilling tolerance,[58] reducing BF development in mango.

Interestingly, the effect of 1-MCP on BF in both 'Tommy Atkins' and 'Keitt' mangoes was not strictly dose-dependent, as low and intermediate concentrations were generally as effective as higher doses in reducing the disorder. This indicates that low 1-MCP concentrations (200 nL $L^{-1}$) may be sufficient to achieve meaningful reductions in BF incidence and severity, which has practical implications for postharvest management strategies. Taken together, these findings indicate that 1-MCP has strong potential as a postharvest treatment to mitigate BF in mangoes, primarily by suppressing ethylene-mediated ripening processes and modulating AOX. However, given that the efficacy of 1-MCP may depend strongly on the cultivar, harvest maturity stage, and storage conditions,[42] further studies are warranted to optimize dose, timing, and integration with other postharvest technologies to ensure consistent efficacy across different production systems.

Considering the probability of BF incidence in fruit from orchards with a history of high BF occurrence, this study indicated that mangoes harvested at early maturity and stored at low temperatures (≤10 °C) without 1-MCP treatment exhibited the highest probability of BF incidence (++BF). However, this incidence could be markedly reduced by 1-MCP application (−BF) or by storing the fruit at a higher temperature of 12.5 °C (−BF) (Fig. 8). Conversely, harvesting fruit at more advanced maturity stages tended to decrease BF incidence in untreated fruit (+BF) and enhance the suppressive effects of both 1-MCP treatment (−BF) and higher storage temperature (12.5 °C; −BF) on BF incidence during long-term cold storage or transportation (Fig. 8).

## CONCLUSION

Black flesh is a CI symptom that is strongly aggravated by early harvest and storage at or below 10 °C, whereas storage at 12.5 °C completely prevents the disorder in all evaluated mango cultivars.

The application of 1-MCP mitigated BF incidence and severity in both 'Tommy Atkins' and 'Keitt' mangoes, suppressing ethylene production, reducing respiration rate, maintaining flesh firmness, and preserving the antioxidant system during cold storage and subsequent shelf life.

Among the 1-MCP concentrations tested, 200 nL $L^{-1}$ proved sufficient to reduce BF incidence and severity, and to maintain higher flesh firmness during long-term storage.

1-Methylcyclopropene represents a promising postharvest strategy to alleviate chilling-related disorders and extend the postharvest life of mangoes without compromising fruit quality.

## AUTHOR CONTRIBUTIONS

**Bruna Parente de Carvalho Pires:** writing – review and editing, writing – original draft, visualization, validation, methodology, investigation, formal analysis, data curation, conceptualization. **João Claudio Vilvert:** writing – review and editing, writing – original draft, visualization, validation, methodology, investigation, formal analysis, data curation, conceptualization. **Nilo Ricardo Corrêa de Mello Júnior:** validation, methodology, investigation, formal analysis. **Sandy Raiele Sena Monteiro:**


© 2026 The Author(s).


10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

validation, methodology, investigation, formal analysis. **Mikaele de Souza Santos:** validation, methodology, investigation, formal analysis. **Ester Silva Regis:** validation, methodology, investigation, formal analysis. **Jeffrey K. Brecht:** writing – review and editing, writing – original draft, visualization, methodology, conceptualization. **Sergio Tonetto de Freitas:** writing – review and editing, writing – original draft, visualization, validation, supervision, resources, project administration, methodology, investigation, funding acquisition, data curation, conceptualization. **Jackson Teixeira Lobo:** validation, methodology, investigation.

## ACKNOWLEDGEMENTS

The authors wish to thank the Fundação de Amparo à Ciência e Tecnologia do Estado de Pernambuco (FACEPE) for the research scholarship awarded to the first and second contributors (BFT-0055-5.01/23 and BFP-0239-24-76006), as well as the funding support from FACEPE (APQ-1046-5.01/22) and the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) for the Research Productivity Fellowship (grant number 305222/2023-6) awarded to the corresponding author. They also wish to express their gratitude for the financial support received from CNPq through the National Institutes of Science and Technology Program (grant numbers 465768/2014-8 and 408855/2024-0), as well as the mango companies Agropecuária Roriz Dantas, Argofruta Comercial Exportadora Ltda, Ibacem Agrícola, Comércio e Exportação Ltda, and Muranaka Com. Imp. Exp. Ltda for supplying the fruit used in the study, and AgroFresh, Inc. for supplying the 1-methylcyclopropene and the materials required in the study. The article processing fee for the publication of this research was funded by the Coordination for the Improvement of Higher Education Personnel – CAPES (identifier ROR: 00x0ma614). For open access purposes, the authors have assigned a Creative Commons CC BY license to any accepted version of the article. The Article Processing Charge for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614).

## DATA AVAILABILITY STATEMENT

Data will be made available on request.

## CONFLICT OF INTEREST

The authors declare no conflict of interest.

## REFERENCES

1 Zheng B, Shahzaib M, Mi Z, Jiang X, Zhu Y, Cong H *et al*., Electron beam irradiation significantly improves post-harvest fruit quality and shelf-life of mango (*Mangifera indica* L.). *Postharvest Biol Technol* **228**: 113647 (2025). https://doi.org/10.1016/j.postharvbio.2025.113647.

2 Santos LF, de Souza TA, Vilvert JC, da Silva Alves J, de Carvalho Pires BP, Fraga Junior LS *et al*., Anaerobic compensation point can effectively extend 'Palmer' mango shelf-life in CA storage. *Postharvest Biol Technol* **231**:113951 (2026). https://doi.org/10.1016/j.postharvbio.2025.113951.

3 Brazilian Institute of Geography and Statistics, *Mango production* Brazilian Government (2025). Available: https://www.ibge.gov.br/explica/producao-agropecuaria/manga/br/ [24 May 2025].

4 Costa CSR, de Lima MAC, Lima Neto FP, da Silva Costa AE, Vilvert JC, Martins LSS *et al*., Genetic parameters and selection of mango genotypes using the FAI-BLUP multitrait index. *Sci Hortic* **317**:112049 (2023). https://doi.org/10.1016/j.scienta.2023.112049.

5 Vilvert JC, de Freitas ST, Ferreira MAR, Costa CSR, Leite RHL, dos Santos FKG *et al*., Preservation of quality and bioactive compounds in mangoes using chitosan-graphene-oxide-based biodegradable packaging. *Horticulturae* **9**:1145 (2023). https://doi.org/10.3390/horticulturae9101145.

6 Shahzad F, Doron M, Sargent SA and Brecht JK, Development and feasibility of an approach utilizing modified atmosphere packaging (MAP) and ethylene scrubbing to allow extended international transport and marketing of tree-ripe mangos. *Postharvest Biol Technol* **225**:113488 (2025). https://doi.org/10.1016/j.postharvbio.2025.113488.

7 Kist BB, Santos CE, Carvalho C and Beling RR, *Brazilian Horti & Fruit Yearbook*, Vol. **2024**. Editora Gazeta Santa Cruz, Santa Cruz do Sul (2024).

8 Kumar N, Pratibha Upadhyay A, Trajkovska Petkoska A, Gniewosz M and Kieliszek M, Extending the shelf life of mango (*Mangifera indica* L.) fruits by using edible coating based on xanthan gum and pomegranate peel extract. *J Food Meas Charact* **17**:1300–1308 (2023). https://doi.org/10.1007/s11694-022-01706-6.

9 Huang T, Liu G, Zhu L, Liu J, Xiang Y, Xu X *et al*., Mitigation of chilling injury in mango fruit by methyl jasmonate is associated with regulation of antioxidant capacity and energy homeostasis. *Postharvest Biol Technol* **211**:112801 (2024). https://doi.org/10.1016/j.postharvbio.2024.112801.

10 Shivashankar KS, Geetha GA, Ravishankar KV, Rao DVS, Pavithra KCG and Roy TK, Temperature gradient storage induced biochemical and molecular changes in mango (*Mangifera indica* L. cv. Alphonso) fruits. *Acta Physiol Plant* **44**:46 (2022). https://doi.org/10.1007/s11738-022-03380-z.

11 Sati H, Pareek S, Brecht JK and Yahia EM, Mechanisms of chilling injury and technologies to mitigate it in mango fruit: a review. *Food Rev Int* **42**:432–457 (2026). https://doi.org/10.1080/87559129.2025.2487681.

12 Brecht JK, Schaffer B, Crane J, Li Y and Vargas A, *Mango internal discoloration ("Cutting Black" or "Corte Negro"). Final Report to the National Mango Board* National Mango Board (2019). Available: https://www.mango.org/wp-content/uploads/2025/02/Mango-Internal-Discoloration-Final-Report_Feb_3_2019.pdf [11 April 2026].

13 Brecht JK, Mango, in *Postharvest Physiological Disorders in Fruit and Vegetables*, ed. by Freitas ST and Pareek S. CRC Press, Boca Raton, pp. 443–466 (2019).

14 Ullah MA, Kiloes AM, Aziz AA and Joyce DC, Impact of factors contributing to internal disorders of mango (*Mangifera indica* L.) fruit – a systematic literature review. *Sci Hortic* **331**:113150 (2024). https://doi.org/10.1016/j.scienta.2024.113150.

15 Mogollón R, Contreras C, da Silva Neta ML, Marques EJN, Zoffoli JP and de Freitas ST, Non-destructive prediction and detection of internal physiological disorders in 'Keitt' mango using a hand-held Vis-NIR spectrometer. *Postharvest Biol Technol* **167**:111251 (2020). https://doi.org/10.1016/j.postharvbio.2020.111251.

16 Vega-Alvarez M, Salazar-Salas NY, López-Angulo G, Pineda-Hidalgo KV, López-López ME, Vega-García MO *et al*., Metabolomic changes in mango fruit peel associated with chilling injury tolerance induced by quarantine hot water treatment. *Postharvest Biol Technol* **169**: 111299 (2020). https://doi.org/10.1016/j.postharvbio.2020.111299.

17 Chai J, Li J, Li L, Liao B and Liu Z, 1-MCP application in 'Xuxiang' kiwifruit at appropriate firmness avoids hard-core and prolongs the shelf life. *Postharvest Biol Technol* **206**:112549 (2023). https://doi.org/10.1016/j.postharvbio.2023.112549.

18 Ghimire U, Abeli P, Brecht JK, Pliakoni E and Liu T, Unique molecular mechanisms revealed for the effects of temperature, CA, ethylene exposure, and 1-MCP on postharvest senescence of broccoli. *Postharvest Biol Technol* **213**:112919 (2024). https://doi.org/10.1016/j.postharvbio.2024.112919.

19 Zhou H, Ye Z, Wang L, Zhang S, Yuan Z, Su M *et al*., 1-MCP regulates taste development in cold-stored peach fruit through modulation of sugar, organic acid, and polyphenolic metabolism. *Postharvest Biol Technol* **225**:113518 (2025). https://doi.org/10.1016/j.postharvbio.2025.113518.

20 Watkins CB, The use of 1-methylcyclopropene (1-MCP) on fruits and vegetables. *Biotechnol Adv* **24**:389–409 (2006). https://doi.org/10.1016/j.biotechadv.2006.01.005.

21 Wang B, Wang J, Feng X, Lin L, Zhao Y and Jiang W, Effects of 1-MCP and exogenous ethylene on fruit ripening and antioxidants in stored mango. *Plant Growth Regul* **57**:185–192 (2009). https://doi.org/10.1007/s10725-008-9335-y.

22 Yuan F, Wang C, Yi P, Li L, Wu G, Huang F *et al*., The effects of combined 1-methylcyclopropene and melatonin treatment on the



10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

quality characteristics and active oxygen metabolism of mango fruit during storage. *Foods* **12**:1979 (2023). https://doi.org/10.3390/foods12101979.
23 Shikwambana K, Mathaba N and Mafeo TP, The physiological effect of fruit maturity and 1-methylcyclopropene on 'Hass' avocado fruit exocarp colour and chilling injury during ripening. *J Hortic Postharvest Res* **6**:55–76 (2023). https://doi.org/10.22077/jhpr.2022.5538.1287.
24 Zhang W, Zhao H, Jiang H, Xu Y, Cao J and Jiang W, Multiple 1-MCP treatment more effectively alleviated postharvest nectarine chilling injury than conventional one-time 1-MCP treatment by regulating ROS and energy metabolism. *Food Chem* **330**:127256 (2020). https://doi.org/10.1016/j.foodchem.2020.127256.
25 He J, Feng Y, Cheng Y, Wang S, Fu Y and Guan J, The influence of 1-MCP on the fruit quality and flesh browning of 'red Fuji' apple after long-term cold storage. *J Appl Bot Food Qual* **94**:132–139 (2021). https://doi.org/10.5073/JABFQ.2021.094.016.
26 Atoo GH, Ubwa ST, Alakali J, Anhwange BA and Amua QM, Effect of 1-methylcyclopropene concentration, storage temperature and packaging on the postharvest quality of mango (*Mangifera indica* L.) fruit cv. Broken and Dausha. *Int J Innovative Sci Res Tech* **7**:978–998 (2022). https://doi.org/10.5281/zenodo.6820268.
27 Li L, Li C, Sun J, Sheng J, Zhou Z, Xin M *et al*., The effects of 1-methylcyclopropene in the regulation of antioxidative system and softening of mango fruit during storage. *J Food Qual* **2020**:1–11 (2020). https://doi.org/10.1155/2020/6090354.
28 Alvares CA, Stape JL, Sentelhas PC, Gonçalves JDM and Sparovek G, Köppen's climate classification map for Brazil. *Meteorol Z* **22**:711–728 (2013). https://doi.org/10.1127/0941-2948/2013/0507.
29 National Mango Board, *Mango maturity & ripeness guide* National Mango Board (2017). Available: https://www.mango.org/wp-content/uploads/2017/10/Mango_Maturity_And_Ripeness_Guide.pdf [1 May 2025].
30 Ding Z, Tian S, Zheng X, Zhou Z and Xu Y, Responses of reactive oxygen metabolism and quality in mango fruit to exogenous oxalic acid or salicylic acid under chilling temperature stress. *Physiol Plant* **130**:112–121 (2007). https://doi.org/10.1111/j.1399-3054.2007.00893.x.
31 Li P, Zheng X, Liu Y and Zhu Y, Pre-storage application of oxalic acid alleviates chilling injury in mango fruit by modulating proline metabolism and energy status under chilling stress. *Food Chem* **142**:72–78 (2014). https://doi.org/10.1016/j.foodchem.2013.06.132.
32 Obanda M, Owuor PO and Taylor SJ, Flavanol composition and caffeine content of green leaf as quality potential indicators of Kenyan black teas. *J Sci Food Agric* **74**:209–215 (1997). https://doi.org/10.1002/(SICI)1097-0010(199706)74:2%3C209::AID-JSFA789%3E3.0.CO;2-4.
33 Re R, Pellegrini N, Proteggente A, Pannala A, Yang M and Rice-Evans C, Antioxidant activity applying an improved ABTS radical cation decolorization assay. *Free Radic Biol Med* **26**:1231–1237 (1999). https://doi.org/10.1016/S0891-5849(98)00315-3.
34 Kim DO, Jeong SW and Lee CY, Antioxidant capacity of phenolic phytochemicals from various cultivars of plums. *Food Chem* **81**:321–326 (2002). https://doi.org/10.1016/S0308-8146(02)00040-2.
35 Mora AA, Vega PA and Téliz OD, Enfermedades del mango, in *El mango y su manejo integrado en Michoacán*, ed. by Téliz D and Grupo Interdisciplinario de Investigación en Mango. Colegio de Postgraduados en Montecillo, Montecill, Texcoco, México, pp. 18–44 (1998).
36 Acosta-Ramos M, Nieto-Ángel D, Nieto-Ángel R, Mena-Nevaréz G, Téliz D and Vaquera-Hueta H, Presencia de la pulpa negra en frutos de mango 'Haden' durante postcosecha. *Rev Chapingo Ser Hortic* **7**:131–141 (2001). https://doi.org/10.5154/r.rchsh.2000.11.080.
37 Medlicott AP, Sigrist JMM and Sy O, Ripening of mangoes following low temperature storage. *J Amer Soc Hortic Sci* **115**:430–434 (1990). https://doi.org/10.21273/JASHS.115.3.430.
38 Mohammed M and Brecht JK, Influence of ethylene treatments on ripening and chilling injury alleviation in mango (cv. Palmer) fruit. *Acta Hortic* **509**:437–446 (2000). https://doi.org/10.17660/ActaHortic.2000.509.51.
39 Mohammed M and Brecht JK, Reduction of chilling injury in 'Tommy Atkins' mangoes during ripening. *Sci Hortic* **95**:297–308 (2002). https://doi.org/10.1016/S0304-4238(02)00041-9.
40 Alves JS, Parente de Carvalho Pires B, Ferreira dos Santos L, da Silva Ribeiro T, Walsh KB, Akio Kido E *et al*., non-destructive detection of current internal disorders and prediction of future appearance in mango fruit using portable Vis-NIR spectroscopy. *Horticulturae* **11**:759 (2025). https://doi.org/10.3390/horticulturae11070759.
41 Chang LY and Brecht JK, Early chilling injury symptom development and recovery of mature green banana: involvement of ethylene under different severities of chilling stress. *Horticulturae* **10**:1050 (2024). https://doi.org/10.3390/horticulturae10101050.
42 Zhang J, Ma Y, Dong C, Terry LA, Watkins CB, Yu Z *et al*., Meta-analysis of the effects of 1-methylcyclopropene (1-MCP) treatment on climacteric fruit ripening. *Hortic Res* **7**:208 (2020). https://doi.org/10.1038/s41438-020-00405-x.
43 Yang X, Song J, Campbell-Palmer L, Fillmore S and Zhang Z, Effect of ethylene and 1-MCP on expression of genes involved in ethylene biosynthesis and perception during ripening of apple fruit. *Postharvest Biol Technol* **78**:55–66 (2013). https://doi.org/10.1016/j.postharvbio.2012.11.012.
44 Satekge TK and Magwaza LS, Postharvest application of 1-methylcyclopropene (1-MCP) on climacteric fruits: factors affecting efficacy. *Int J Fruit Sci* **22**:595–607 (2022). https://doi.org/10.1080/15538362.2022.2085231.
45 Chang LY, Sargent SA, Kim J and Brecht JK, Delaying ripening using 1-MCP reveals chilling injury symptom development at the putative chilling threshold temperature for mature green banana. *Front Plant Sci* **13**:966789 (2022). https://doi.org/10.3389/fpls.2022.966789.
46 Pachón YV, Balaguera-López HE and Florez-Velasco N, Postharvest behavior and chilling injury in avocado (*Persea americana* Mill) fruit cv. Hass treated with 1-methylcyclopropene, ethylene, and intermittent warming. *Rev Fac Nac Agron Medellín* **75**:9895–9907 (2022). https://doi.org/10.15446/rfnam.v75n2.98741.
47 Razzaq K, Singh Z, Khan AS, Khan SAKU and Ullah S, Role of 1-MCP in regulating 'Kensington pride' mango fruit softening and ripening. *Plant Growth Regul* **78**:401–411 (2016). https://doi.org/10.1007/s10725-015-0101-7.
48 Guan F, Shi B, Yang X, Xie Y and Wan X, Transcriptomic and metabolomic analyses reveal the impact of 1-methylcyclopropene on bitter gourd fruit storage. *Postharvest Biol Technol* **230**:113719 (2025). https://doi.org/10.1016/j.postharvbio.2025.113719.
49 Li X, Xiong T, Zhu Q, Zhou Y, Lei Q, Lu H *et al*., Combination of 1-MCP and modified atmosphere packaging (MAP) maintains banana fruit quality under high temperature storage by improving antioxidant system and cell wall structure. *Postharvest Biol Technol* **198**:112265 (2023). https://doi.org/10.1016/j.postharvbio.2023.112265.
50 Tatsuki M, Hayama H and Nakamura Y, Apple ethylene receptor protein concentrations are affected by ethylene, and differ in cultivars that have different storage life. *Planta* **230**:407–417 (2009). https://doi.org/10.1007/s00425-009-0953-z.
51 Tatsuki M and Endo A, Analyses of expression patterns of ethylene receptor genes in apple (*Malus domestica* Borkh.) fruit treated with or without 1-methylcyclopropene (1-MCP). *J Japan Soc Hort Sci* **75**:481–487 (2006).
52 Mata CI, Fabre B and Parsons HT, Hertog MLATM, Raemdonck GV, Baggerman G, Poel BV, Lilley KS and Nicolaï BM, ethylene receptors, CTRs and EIN2 target protein identification and quantification through parallel reaction monitoring during tomato fruit ripening. *Front Plant Sci* **9**:1–16 (2018). https://doi.org/10.3389/fpls.2018.01626.
53 Zhao Q, Zhou X, Chen Q, Yang R, Li Y, Zhao J *et al*., ER-related E2-E3 ubiquitin enzyme pair regulates ethylene response by modulating the turnover of ethylene receptors. *Nat Commun* **16**:5937 (2025). https://doi.org/10.1038/s41467-025-61066-9.
54 Chomba G, Ambuko J, Onyango C and Ouko JR, Effect of different 1-methylcyclopropene formulations and dosing on the ripening profile of Tommy Atkins mango fruits. *Front Hortic* **4**:1509989 (2025). https://doi.org/10.3389/fhort.2025.1509989.
55 Lima LCO, Chitarra AB and Chitarra MIF, Changes in amylase activity starch and sugars contents in mango fruits pulp cv. Tommy Atkins with spongy tissue. *Braz Arch Biol Technol* **44**:59–62 (2001). https://doi.org/10.1590/S1516-89132001000100008.
56 Guo S, Li T, Wu C, Fan G, Wang H and Shen D, Melatonin and 1-methylcyclopropene treatments on delay senescence of apricots during postharvest cold storage by enhancing antioxidant system activity. *J Food Process Preserv* **45**:e15863 (2021). https://doi.org/10.1111/jfpp.15863.
57 Rey F, Zacarías L and Rodrigo MJ, Carotenoids, vitamin C, and antioxidant capacity in the peel of mandarin fruit in relation to the susceptibility to chilling injury during postharvest cold storage. *Antioxidants* **9**:1296 (2020). https://doi.org/10.3390/antiox9121296.
58 Valenzuela JL, Manzano S, Palma F, Carvajal F, Garrido D and Jamilena M, Oxidative stress associated with chilling injury in immature fruit: postharvest technological and biotechnological solutions. *Int J Mol Sci* **18**:1467 (2017). https://doi.org/10.3390/ijms18071467.



10970010, 0, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.70681 by Capes, Wiley Online Library on [27/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License